Machine Learning — Day 1

Fundamentals & Linear Regression

GitHub-ready notes covering Machine Learning fundamentals, its types, supervised learning, regression and classification, and Linear Regression.

1. What is Machine Learning?

Machine Learning (ML) is a subset of Artificial Intelligence where computers learn patterns from data and use those patterns to make predictions or decisions without being explicitly programmed for every possible case.

Traditional Programming

```
Input + Rules -> Output
```

Machine Learning

```
Input Data + Expected Output
            |
      ML Algorithm
            |
        Learned Model
```

Example: Given hours studied and exam scores, a model can learn their relationship and predict the expected score for a new student.

2. AI vs ML vs Deep Learning

```
Artificial Intelligence
|
+-- Machine Learning
|   +-- Linear Regression
|   +-- Decision Trees
|   +-- SVM
|   +-- K-Means
|
+-- Deep Learning
    +-- Neural Networks
    +-- CNN
    +-- RNN
    +-- Transformers
```

AI: The broad field of creating systems capable of tasks that normally require human intelligence.

ML: A subset of AI where systems learn patterns from data.

Deep Learning: A subset of ML primarily based on multi-layer neural networks.

3. Why Do We Need Machine Learning?

Rule-based programming works well when rules are known. But tasks such as recognizing cats and dogs are difficult to solve by manually writing every possible rule. ML learns useful patterns from examples.

```
Images + Labels
      |
Machine Learning
      |
Learned Patterns
      |
New Image -> Prediction
```

4. Types of Machine Learning

```
Machine Learning
|
+-- Supervised Learning
+-- Unsupervised Learning
+-- Semi-Supervised Learning
+-- Reinforcement Learning
```

4.1 Supervised Learning

The model learns from labeled data: **Input -> Correct Output**. We can think of this as X -> Y.

Regression: predicts continuous numerical values such as house price, salary, temperature, or exam score.

Classification: predicts categories such as spam/not spam, fraud/genuine, or cat/dog.

4.2 Unsupervised Learning

The model receives data without predefined labels and tries to discover hidden structure or patterns.

Applications include customer segmentation, document grouping, market analysis, and anomaly detection. Common techniques include K-Means, hierarchical clustering, and PCA.

4.3 Reinforcement Learning

An agent interacts with an environment, takes actions, receives rewards or penalties, and learns to improve future decisions.

```
Agent -> Action -> Environment
Agent <- Reward <- Environment
```

Applications include game-playing AI, robotics, autonomous systems, and decision-making.

4.4 Semi-Supervised Learning

Combines a small amount of labeled data with a large amount of unlabeled data. It is useful when labeling data is expensive or time-consuming.

5. Important ML Terminology

Term	Meaning
Dataset	Collection of examples/data
Feature	Input variable used by the model
Target / Label	Output value the model tries to predict
Sample / Instance	One individual example or row
Model	Mathematical representation of a learned pattern
Parameter	Value learned by the model
Prediction	Output produced by the model
Error	Difference between actual and predicted value
Loss	Mathematical measure of model error
Training	Process of learning model parameters

6. Linear Regression

Linear Regression is a supervised learning algorithm used primarily to predict continuous numerical values. Its goal is to find the best-fitting straight line representing the relationship between input and output.

Example Dataset

Hours Studied	Exam Score
1	35
2	40
3	50
4	55
5	65
6	70
7	80

6.1 Linear Regression Equation

The equation of a straight line is **y = mx + c**. In Machine Learning, it is commonly written as:

y_hat = b0 + b1*x

x = input feature, **y_hat** = predicted output, **b0** = intercept, **b1** = coefficient/slope.

6.2 Understanding the Slope

Suppose the model is **y_hat = 28 + 7*x**. Here b0 = 28 and b1 = 7. The coefficient 7 means that for every 1-unit increase in x, the predicted output increases by 7 units.

6.3 Understanding the Intercept

The intercept b0 is the predicted value when x = 0. It is a model parameter and may or may not have a meaningful real-world interpretation.

6.4 Making a Prediction

For **y_hat = 28 + 7*x** and x = 6:

y_hat = 28 + 7(6) = 70

The model predicts an exam score of 70.

7. How Does the Model Learn?

We do not manually provide the final values of b0 and b1. The algorithm learns them from the training data.

```
Training Data
      |
Try parameters
      |
Make predictions
      |
Compare predictions with actual values
      |
Measure error
      |
Improve parameters
      |
Repeat
```

8. Prediction Error

If the actual score is 80 and the predicted score is 70:

Error = y - y_hat = 80 - 70 = 10

For an individual observation, e_i = y_i - y_hat_i. This difference is also called the residual.

9. Mean Squared Error (MSE)

MSE is a common loss function for Linear Regression:

MSE = (1/n) * SUM((y_i - y_hat_i)^2)

The process is: calculate each prediction error, square it, then average the squared errors.

Squaring prevents positive and negative errors from cancelling out and gives larger errors greater influence.

Example

```
Errors: 5, -3, 8, -2

5^2  = 25
(-3)^2 = 9
8^2  = 64
(-2)^2 = 4

MSE = (25 + 9 + 64 + 4) / 4
    = 25.5
```

10. Objective of Linear Regression

The goal is to find values of b0 and b1 that minimize the prediction error, commonly measured using MSE.

```
Model:       y_hat = b0 + b1*x
Loss:        MSE
Objective:   Minimize MSE
Parameters:  b0 and b1
```

11. Gradient Descent - Basic Idea

Gradient Descent is one common optimization method. It repeatedly adjusts model parameters in a direction that reduces the loss.

```
Parameters
    |
Predictions
    |
Loss
    |
Gradient
    |
Update Parameters
    |
Repeat
```

The detailed mathematics of Gradient Descent will be covered in the next lesson.

12. Simple vs Multiple Linear Regression

Simple Linear Regression

Uses one feature: **y_hat = b0 + b1*x**. Example: Hours Studied -> Exam Score.

Multiple Linear Regression

Uses multiple features: **y_hat = b0 + b1*x1 + b2*x2 + ... + bn*xn**.

Example features: hours studied, attendance, previous score, assignments, and sleep.

13. Model vs Loss vs Optimization

Component	Purpose
Model	Makes predictions
Loss Function	Measures prediction error
Optimization Algorithm	Finds better parameter values

14. Complete Linear Regression Pipeline

```
Training Data
      |
Select Features
      |
Linear Model: y_hat = b0 + b1*x
      |
Prediction
      |
Compare with Actual
      |
Calculate MSE
      |
Optimization
      |
Update Parameters
      |
Repeat
      |
Trained Model
      |
New / Unseen Data
      |
Prediction
```

15. Today's Revision Checklist

[ ] What is Machine Learning?

[ ] AI vs ML vs Deep Learning

[ ] Why ML is needed

[ ] Types of Machine Learning

[ ] Supervised vs Unsupervised vs Reinforcement Learning

[ ] Regression vs Classification

[ ] Dataset, feature, target, sample

[ ] Linear Regression

[ ] Simple Linear Regression

[ ] Intercept and slope

[ ] Prediction

[ ] Prediction error and residual

[ ] Mean Squared Error

[ ] Model vs loss vs optimization

[ ] Basic idea of Gradient Descent

[ ] Multiple Linear Regression

[ ] Complete ML pipeline

16. Next Topic

```
Linear Regression
      |
Cost/Loss Function in Depth
      |
Gradient Descent
      |
Learning Rate
      |
Gradient Descent for b0 and b1
      |
Normal Equation
      |
Train/Test Split
      |
MAE, MSE, RMSE, R2
      |
Overfitting / Underfitting
      |
Regularization
      |
Ridge / Lasso
```